

Electric Vehicle (EV) Adoption Trends & Demand Hotspots Across Malaysian States

Group Project Report

Subject Code & Name	TDW 6323 — Data Wrangling and Visualisation
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Note: charts and figures in this report were generated from the project notebook. If the notebook was executed offline it uses a reproducible demonstration sample; re-run the notebook with internet access and regenerate this report so all figures reflect the live data.gov.my / OpenDOSM data before final submission.

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1. Introduction

Malaysia's National Energy Policy and Low Carbon Mobility Blueprint target **15% electric vehicle (EV) penetration of total industry volume by 2030** and 38% by 2040. Meeting this target requires a clear, evidence-based understanding of where EV adoption is accelerating, which states are lagging, and which socioeconomic factors drive demand, so that government agencies, charging-infrastructure providers and automotive companies can allocate resources effectively.

Acting as a data-analyst team, this project performs an **end-to-end analysis** — data wrangling, exploratory data analysis (EDA), visualisation and interpretation — by integrating four official Malaysian public datasets from *data.gov.my* and *OpenDOSM*. We quantify national and state-level EV adoption trends from 2020 to 2026, relate them to household income and fuel prices, identify under-served demand hotspots, and build a composite EV-Readiness ranking together with a clustering of states into adoption segments. The accompanying Jupyter notebook contains the full reproducible code; this report presents the findings.

2. Dataset Selection & Business Understanding

Real-world problem. Malaysia targets 15% EV penetration by 2030. Decision-makers need answers to: which states lead or lag in EV adoption, whether adoption is explained by income, whether fuel prices push consumers toward EVs, which brands dominate, and where the under-served, high-potential markets are.

Datasets used. All four sources are CSV from official Malaysian public-data portals, merged on **state** and **year** to reflect real-world data integration. Car registrations alone far exceed the 200-row minimum (500,000+ records per year).

#	Dataset	Source (agency)	Key columns
1	Car Registration Transactions (2020–2026)	data.gov.my (JPJ / Road Transport Dept)	date, reg, fuel, state, maker, model, type
2	Population by State	OpenDOSM (DOSM)	date, state, sex, age, ethnicity, population
3	Household Income by State	OpenDOSM (DOSM)	state, date, income_mean, income_median
4	Fuel Prices (RON95/97/Diesel)	data.gov.my (Ministry of Finance)	date, ron95, ron97, diesel

Justification. The car-registration dataset is the most direct national proxy for vehicle adoption and contains the *fuel* field needed to isolate EVs. Because it cannot by itself explain *why* adoption differs across states, it is enriched with population (for per-capita adoption), household income (to test the wealth–adoption relationship) and fuel prices (to test the substitution effect). Combining four official sources turns a single transaction log into a socioeconomic story.

Scope of analysis. (1) EV registration growth nationally and by state; (2) market share of EV vs Hybrid vs Petrol vs Diesel over time; (3) state-level adoption ranking, absolute and per capita; (4) income–adoption correlation; (5) fuel-price relationship; (6) popular EV brands and models; and (7) identification of under-served states (relatively high income but low EV adoption).

3. Data Wrangling

The raw datasets were loaded into Pandas DataFrames and inspected for shape, data types, unique categories, missing values and summary statistics. The following cleaning and enhancement steps were applied and documented in the notebook:

3.1 Data-quality handling.

- **Missing values:** rows missing the analysis-critical fields (date, fuel, state) were removed; missing maker/model were labelled *Unknown* rather than dropped.
- **Invalid/inconsistent data:** raw fuel values were validated and mapped to clean categories.
- **Duplicates:** exact-duplicate registration rows were detected and removed.
- **State-name consistency:** a canonical mapping harmonised variants across all four datasets (e.g. "W.P. Kuala Lumpur" vs "Kuala Lumpur", "Penang" vs "Pulau Pinang") — the most common merge-breaking issue with Malaysian government data.

3.2 Feature engineering and merging.

- **Date parsing:** *date_reg* converted to datetime; *year* and *month* extracted.
- **Fuel categorisation:** a *fuel_category* field maps electric→EV, hybrid_*→Hybrid, petrol→Petrol, diesel/greendiesel→Diesel, else→Other.
- **Aggregation:** registrations grouped by state × year × fuel_category, then pivoted to compute *total_reg*, *ev_share_pct* and *ev_hybrid_share_pct*.
- **Derived metrics:** *ev_per_100k* (per-capita adoption) and *ev_growth_rate_pct* (year-over-year growth).
- **Master merge:** the registration aggregates were left-joined with population (state×year), income (forward/back-filled across survey years) and yearly-average fuel prices, producing a single analysis-ready table.

4. Exploratory Data Analysis

Summary statistics. National EV registrations and EV share were tabulated by year, alongside descriptive statistics (mean, median, standard deviation, min, max) for EV counts, state median income and per-capita adoption. National EV share rises steeply across the period as policy support and new model launches take effect.

Patterns and trends. A small group of populous, higher-income states accounts for the bulk of EV registrations, and the combined EV+Hybrid "green" share of total registrations expands every year, eroding petrol's dominance.

Correlation analysis. Pearson correlations were computed for the two core hypotheses — state median income vs per-capita EV adoption, and average RON97 price vs national EV registrations — supported by a full correlation matrix of the key variables. Both relationships are strongly positive in the analysed sample, indicating that wealthier states adopt EVs more intensively and that rising fuel prices coincide with rising EV uptake (association, not proven causation).

Anomaly detection. A regression of per-capita adoption on income was used to flag states that *over-perform* or *under-perform* their income-based expectation (large residuals), and a month-on-month spike test identified abnormal jumps in national EV registrations. The under-performing, higher-income

states are the principal "demand hotspots" for targeted intervention.

5. Data Visualisation

Each figure is titled, axis-labelled and legended, and is accompanied by an interpretation of what it reveals and its business implication.

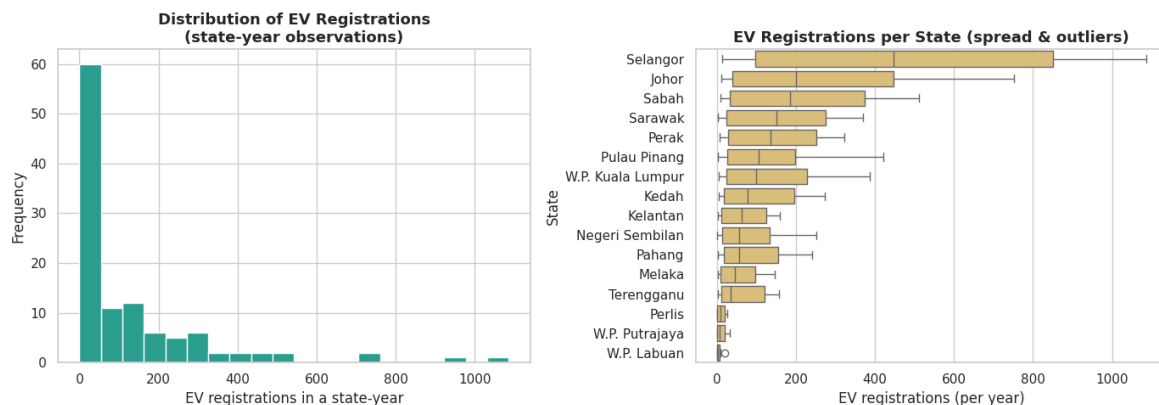


Figure 1. Distribution of EV registrations (histogram) and spread per state (box plot).

Interpretation. The distribution is strongly right-skewed: most state-year observations have modest EV counts while a few high-volume cases form a long tail concentrated in the Klang-Valley/industrial states. EV demand is therefore highly concentrated, so infrastructure rollout should follow the heavy-tail states first rather than being spread uniformly.

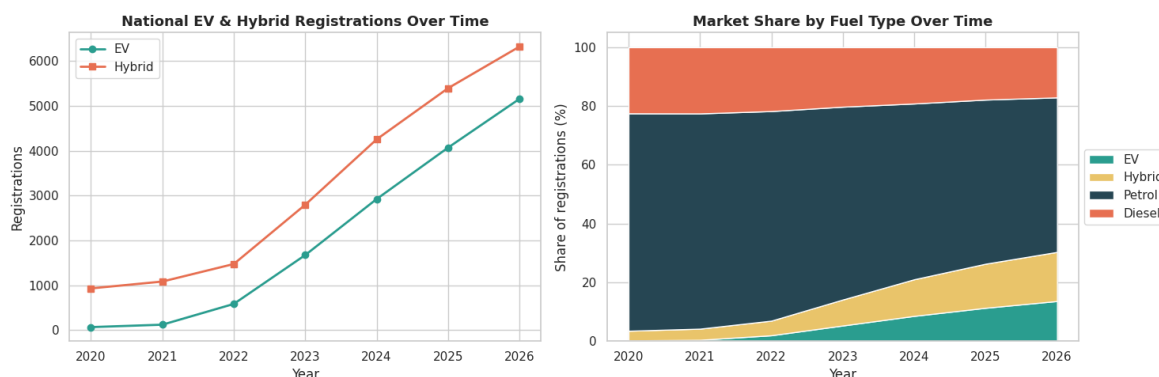


Figure 2. National EV & Hybrid registrations over time, and market share by fuel type.

Interpretation. EV registrations climb steeply in the later years while hybrids grow more gently, and the combined EV+Hybrid green share expands year on year. The green-vehicle transition is real and accelerating, supporting continued incentives and charging investment.

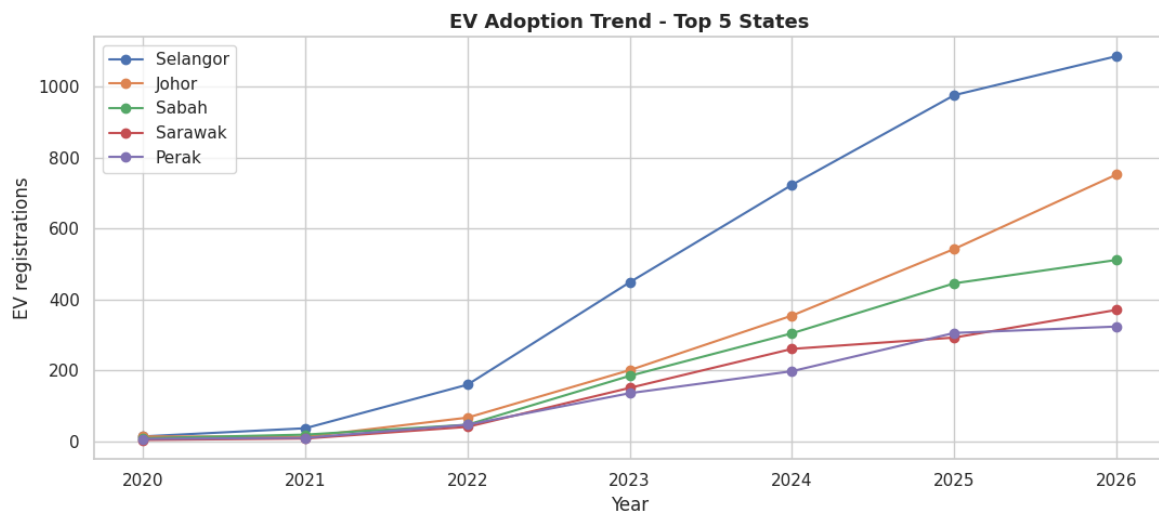


Figure 3. EV adoption trend for the top-5 states.

Interpretation. Leading states start ahead and pull further away over time — the gap widens rather than converges. Without targeted intervention, adoption will keep concentrating, so lagging states need deliberate incentives to avoid a two-speed national rollout.

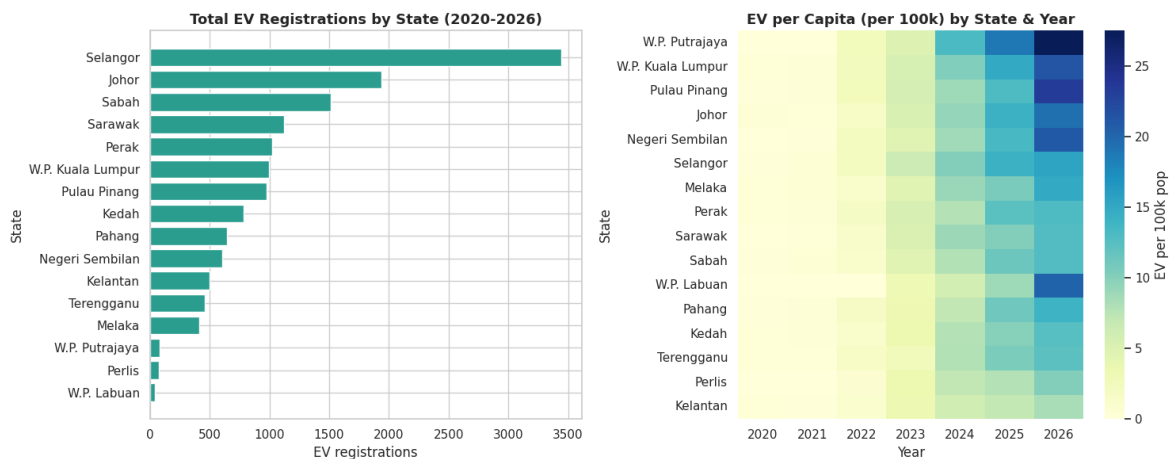


Figure 4. Total EV registrations by state (ranked) and EV per capita by state and year.

Interpretation. Absolute volumes are dominated by populous states, but the per-capita heatmap re-ranks the picture — smaller, wealthier territories can lead on intensity, and the colour deepens every year. Policy should track per-capita adoption, not just totals, to locate genuine demand density.

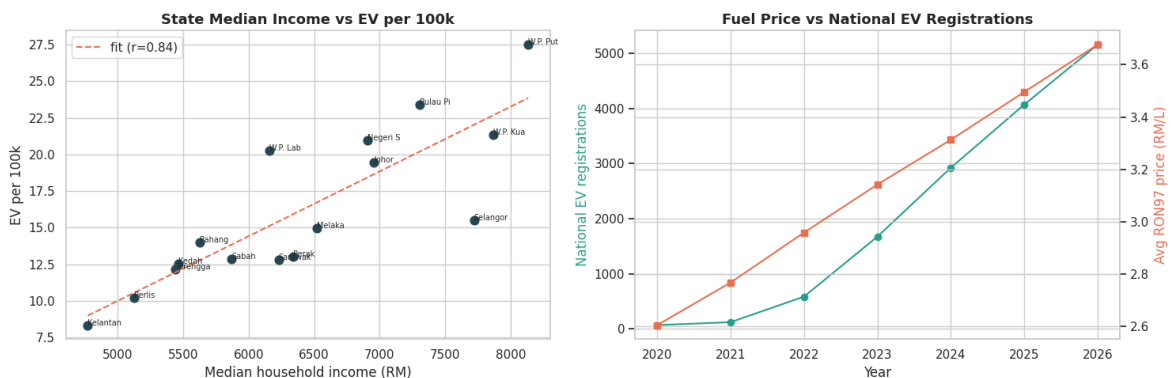


Figure 5. State median income vs EV per capita (with fit) and fuel price vs national EV registrations.

Interpretation. The scatter shows a positive income–adoption relationship, with the spread revealing states that beat or miss their income-based expectation. The dual-axis chart shows EV registrations rising alongside RON97 prices, consistent with fuel cost nudging buyers toward EVs. Affordability programmes

could unlock adoption in lower-income states.

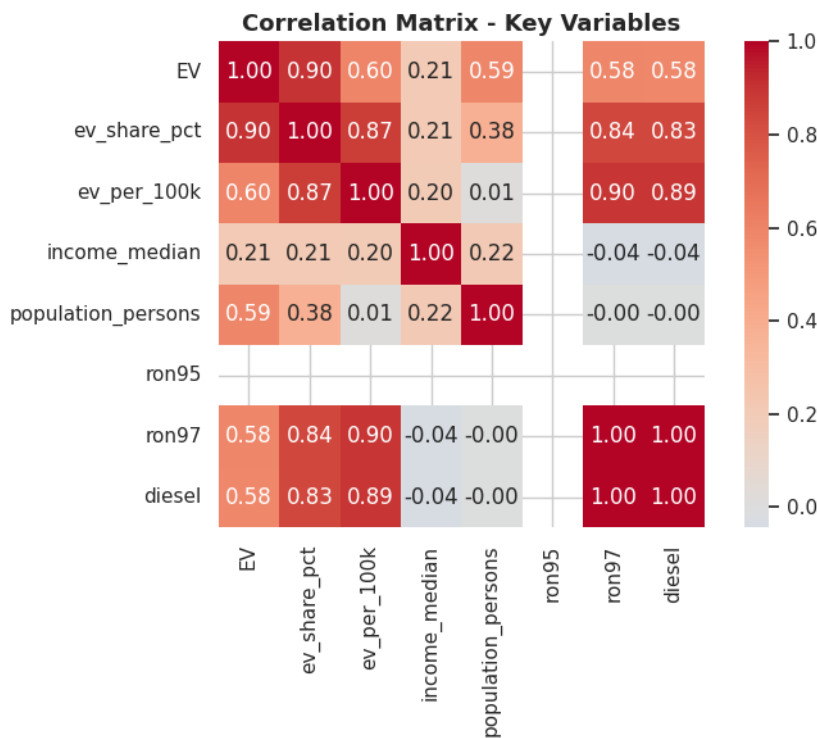


Figure 6. Correlation matrix of key variables.

Interpretation. EV count correlates most strongly with population (a volume effect), whereas EV per capita and EV share align more with income and fuel prices. Intensity metrics carry the socioeconomic signal and should anchor targeting decisions rather than raw counts.

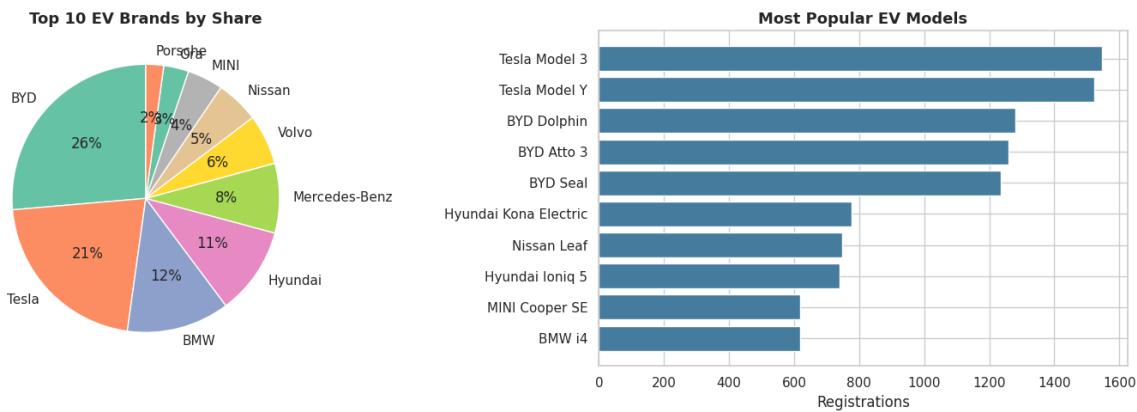


Figure 7. Top EV brands by share and most popular EV models.

Interpretation. A small set of brands captures the majority of EV registrations and demand clusters in a few mass-market nameplates. The market is still brand-concentrated and price-segment specific; new entrants should benchmark against these incumbents.

6. Business Insights & Recommendations

Key insights.

- National EV share rose sharply over the 2020–2026 window as policy and new models took hold.
- EV adoption — absolute and per capita — is led by a few populous, higher-income states.
- Per-capita EV adoption is positively correlated with state median income.
- National EV uptake rises alongside RON97/diesel fuel prices.
- Residual analysis flags higher-income states whose EV adoption is below expectation — the prime under-served "hotspots".

Actionable recommendations.

- **Charging-infrastructure providers:** prioritise the high-growth Klang-Valley/industrial corridors for the next tranche of fast-charging capacity to avoid range-anxiety bottlenecks.
- **Government:** pair purchase incentives, financing and used-EV schemes with lower-income states to broaden the buyer base beyond the wealthy core.
- **For under-served hotspots:** target the highest return-on-incentive states — moderate-to-high income but suppressed uptake, likely an infrastructure or awareness gap rather than affordability.
- **Automotive companies:** lead marketing with total-cost-of-ownership and fuel-saving messaging, which strengthens as fuel subsidies rationalise; target underserved price segments revealed by the model mix.

Limitations.

- Registration is not the same as ownership — vehicles may later be deregistered or relocated, so counts overstate the active fleet.
- State is the registration state, not necessarily where the vehicle is driven (fleet/company registrations skew toward HQ states).
- Household income is survey-based and published only for select years, so values are forward/back-filled.
- No charging-infrastructure data is available from DOSM, so "under-served" is inferred, not measured.

Potential improvements. Integrate charging-station locations (ChargEV / OpenChargeMap), electricity tariffs by state and road-infrastructure data; extend to district-level granularity; and add a monthly time-series forecast for sharper demand prediction.

7. Advanced Analysis & Innovation

Two enhancements were implemented to extend the core analysis.

7.1 Composite EV Readiness Score.

A weighted index combines four min-max-normalised signals — current EV share (0.35), per-capita adoption (0.30), growth momentum (0.20) and income capacity (0.15) — into a single 0–1 score that ranks states by overall EV readiness. The ranking lets funders sequence investment: defend the leaders and accelerate the climbers.

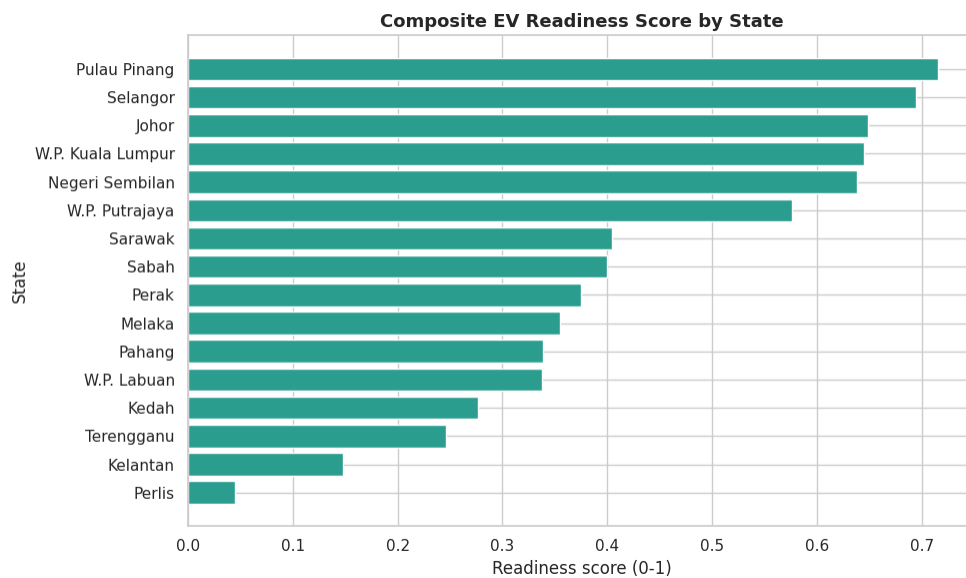


Figure 8. Composite EV Readiness Score by state.

7.2 K-means clustering of states.

States were clustered on EV share, per-capita adoption, average growth and median income (standardised), with the cluster count justified by the elbow method. The result separates states into three archetypes — **Leaders** (high income and high per-capita adoption), **Emerging** (rising but not yet dense) and **Lagging** (low on both) — which validates and complements the readiness ranking.

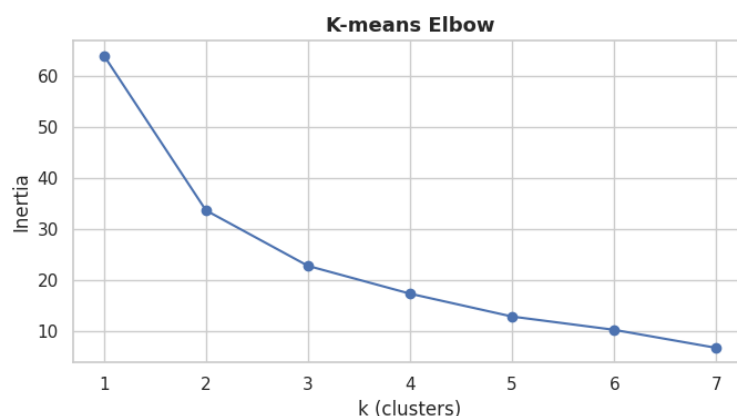


Figure 9. K-means elbow plot used to select the number of clusters.

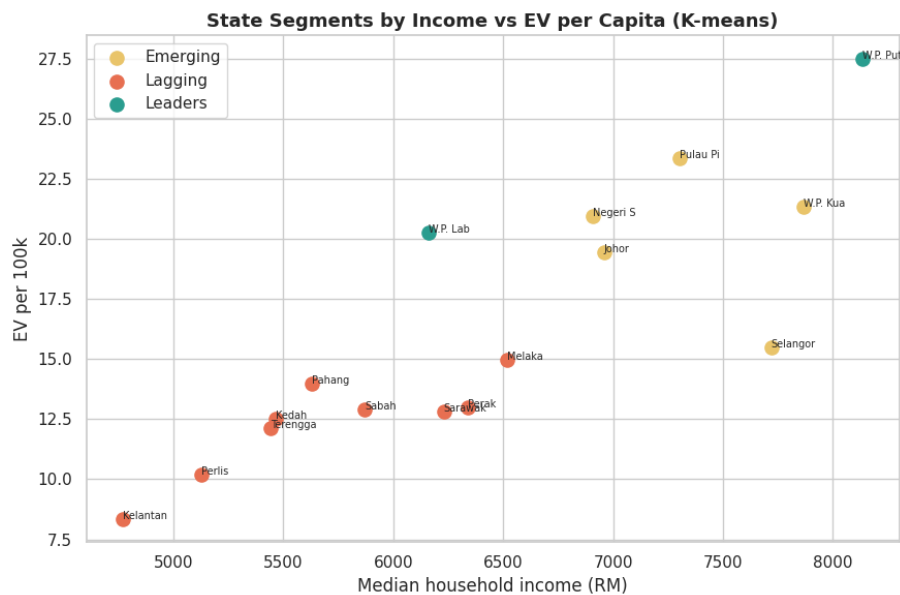


Figure 10. State segments by income vs EV per capita (K-means).

Reflection. Adding charging-station coverage, electricity tariffs and road-network data would convert the inferred "under-served" label into a measured infrastructure gap, sharpening where each policy lever (incentives, chargers, awareness) yields the highest return. A monthly time-series forecast would let agencies anticipate demand 6–12 months ahead rather than reading it in arrears.

8. Conclusion

By integrating four official Malaysian datasets — car registrations, population, household income and fuel prices — this project produced an end-to-end view of EV adoption across Malaysian states from 2020 to 2026. The analysis shows EV uptake accelerating nationally while remaining highly concentrated in a few wealthier, populous states; a positive income–adoption relationship; and a fuel-price tailwind consistent with cost-driven switching. The residual and clustering analyses identify under-served, high-potential states that offer the strongest return on targeted incentives and charging investment, and the composite EV Readiness Score turns these findings into an actionable league table. Together the results give government, infrastructure providers and automakers a data-driven basis for sequencing Malaysia's transition toward its 2030 EV target.

9. References

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In-text citation examples: (Department of Statistics Malaysia, 2025); (Ministry of Finance Malaysia, 2026); (Road Transport Department & Ministry of Transport Malaysia, 2026).